

A Minimality-Preserving Statman Translation

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Abstract

A formula is minimal in a logic L if it belongs to L and no strictly more general formula, in the sense of the substitution preorder, belongs to L . We show that the polarized extension-variable translation S of Tseitin and Statman preserves minimality: for every consistent logic L that proves $\gamma \rightarrow \gamma$ and $S(\alpha)$, if α is minimal in L then so is $S(\alpha)$. This applies to all consistent implicational logics containing **BCI**, to extensions of **FL_e** in the multiplicative language, and to extensions of **FL_{ew}** (in particular of intuitionistic logic) in the full language, and in these cases $L + \alpha = L + S(\alpha)$. For implicative α , $S(\alpha)$ has order at most four. Consequently, every implicative logic over **BCI** axiomatizable by formulas minimal in classical logic is axiomatizable by such formulas of order at most four, and four is optimal. Applied to the counter-example of Nakamura and Matsuda to the Komori–Kashima problem, this yields an explicit order-four axiom of Gödel–Dummett logic minimal in classical logic, answering their Open Problem 2. We also show that the base logic **BCI** cannot be weakened to **CK** or **BK**. The implicational results are formalized in Lean 4.

Keywords: Intermediate logic, Substructural logic, Minimal formulas, Substitution preorder, Axiomatization, Extension variables, Statman translation.

1 Introduction

1.1 The main theorem

Komori [13] studied logics through the *substitution preorder*: $\beta \leq_{\text{sub}} \alpha$ iff α is a substitution instance of β . A formula α is *minimal* in a logic L if $\alpha \in L$ and no strictly more general formula belongs to L . For example, Peirce’s law $((a \rightarrow b) \rightarrow a) \rightarrow a$ is minimal in classical logic **CL**.

The *polarized Statman translation* S (Definition 3.1) replaces every compound subformula occurrence u of α by a fresh variable x_u . It records the meaning of x_u in a premise E_u , a one-way definition whose direction depends on the polarity of u :

$$S(\alpha) = E_{u_1} \rightarrow \cdots \rightarrow E_{u_m} \rightarrow x_{\text{root}}.$$

For Peirce’s law, with x, y, w standing for the whole formula, $(a \rightarrow b) \rightarrow a$ and $a \rightarrow b$,

$$S(((a \rightarrow b) \rightarrow a) \rightarrow a) = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (w \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Each premise E_u contains exactly two connectives. So for implicative α , $S(\alpha)$ has order at most 4 and is computed in linear time.

Theorem 1.1 (Main Theorem). *Let L be a consistent logic that proves $\gamma \rightarrow \gamma$ for every formula γ . If α is minimal in L and $S(\alpha) \in L$, then $S(\alpha)$ is minimal in L .*

The hypothesis $S(\alpha) \in L$ holds for every $\alpha \in L$ as soon as L contains linear implicational logic **BCI** (Corollary 1.2 below).

1.2 Why the theorem holds

A generalization of $S(\alpha)$ can only “forget” some identifications made in $S(\alpha)$. The point is that each such forgetting can be pushed back to α . The proof has two steps.

1. *It suffices to consider one-step generalizations* (Corollary 5.5). A formula in L is minimal iff no formula obtained from it by replacing some occurrences of *one* subformula by a fresh variable z is in L .
2. *Every one-step generalization of $S(\alpha)$ decodes to one of α* (Lemma 5.7). Let α' be such a generalization. We give a substitution θ that maps each (possibly damaged) definition in α' to a formula of the form $\gamma \rightarrow \gamma$, and maps the conclusion to a one-step generalization β of α :

$$\theta\alpha' = (\gamma_1 \rightarrow \gamma_1) \rightarrow \cdots \rightarrow (\gamma_k \rightarrow \gamma_k) \rightarrow \beta.$$

If $\alpha' \in L$, then $\beta \in L$ by modus ponens. This is impossible, because α is minimal.

Only closure under substitution and modus ponens, the formulas $\gamma \rightarrow \gamma$ and consistency are used. So the argument works for any logic. Polarity is needed only to guarantee $S(\alpha) \in L$. For Peirce’s law, a generalization of S that separates the two occurrences of w is decoded in Example 5.8.

1.3 What follows

Corollary 1.2. *Suppose that one of the following holds:*

- (i) *L is an implicational logic containing **BCI** (e.g. **BCI**, **BCK**, **IL**, **LC**, **CL**);*
- (ii) *L is a logic in the language $\{\rightarrow, \cdot, 0, 1\}$ containing the full Lambek calculus with exchange **FL_e**;*
- (iii) *L contains **FL_{ew}** (**FL_e** with weakening), in a language with $\rightarrow$ and some of $\cdot, \wedge, \vee, 0, 1, \perp, \top$ (e.g. **IL** in the full language).*

Then $L + \alpha = L + S(\alpha)$ for every α . If moreover L is consistent and α is minimal in L , then $S(\alpha)$ is minimal in L .

Taking $L = \mathbf{CL}$ for minimality and a base logic $L_0 \supseteq \mathbf{BCI}$ for the axiomatization, we obtain a normal form:

Corollary 1.3 (Order-four normal form). *Let L_0 be an implicational logic containing **BCI** (e.g. **BCI**, **BCK**, **IL**). If $L = L_0 + \Gamma$ with every $\alpha \in \Gamma$ minimal in **CL**, then $L = L_0 + \{S(\alpha) \mid \alpha \in \Gamma\}$, where every $S(\alpha)$ is minimal in **CL** and of order at most 4.*

For the Komori–Kashima problem [13, 12], this has the following consequences.

- Nakamura and Matsuda [14] gave a **CL**-minimal axiom G' of Gödel–Dummett logic **LC** of order 5, and asked whether order 4 suffices [14, Open Problem 2]. $S(G')$ is a **CL**-minimal axiom of **LC** of order 4, so the answer is yes. The formula is displayed in Corollary 4.2.
- The bound 4 is optimal: classical tautologies of order at most 3 are intuitionistic theorems (Proposition 4.1).
- Corollary 1.3 holds over **BCK**, so it also applies to Komori’s original super-**BCK** formulation [14, footnote 1]. For the question which logics are axiomatizable by **CL**-minimal formulas [14, Open Problem 1], one may restrict attention to order 4.

The base logic **BCI** cannot be weakened by dropping B or C (Proposition 6.1). The implicational case of the main theorem and Corollary 1.2(i) are formalized in Lean 4 (Section 7).

1.4 Related work and outline

The translation is not new. Extension variables go back to Tseitin [9]. Statman [15] used them to reduce intuitionistic propositional logic to its implicational fragment (see also Švejdar [8]). One-way definitions chosen by polarity appear in the Plaisted–Greenbaum translation [6] and in Mints’ normal form [5]. Rybakov [7] uses extension variables for rules. Closest to us is Jeřábek [4]. His φ^+ is our $S(\varphi)$ with an extra “ $\wedge 1$ ” in each definition; with it he shows that the substructural hierarchy of Ciabattoni, Galatos and Terui [2] collapses to $\mathcal{N}_3$ over **FL_e**. To our knowledge, the preservation of minimality has not been observed before. The extra “ $\wedge 1$ ” destroys it (Section 6.2).

[Plot: Check the exact forms used in [15], [8], [5].]

The rest of the paper is organized as follows.

- Section 2: preliminaries.
- Section 3: the translation and its basic properties.
- Section 4: the consequences above, with $S(G')$ displayed.
- Section 5: the proof of Theorem 1.1.
- Section 6: supplementary remarks (weaker base logics, additive connectives, premises of order 3).
- Section 7: the formalization.
- Section 8: open problems.

2 Preliminaries

Formulas and logics. We work in a propositional language $\mathcal{L}$ that contains $\rightarrow$ and possibly some of the binary connectives $\cdot$ (fusion), $\wedge$, $\vee$ and some constants (among $\perp, \top, 0, 1$). An *atom* is a variable or a constant. $\alpha \rightarrow \beta \rightarrow \gamma$ stands for $\alpha \rightarrow (\beta \rightarrow \gamma)$. As in [14, Sect. 2], a *logic* is a set L of $\mathcal{L}$ -formulas closed under modus ponens and substitution; $L + F$ is the

smallest logic containing $L \cup F$. L is *consistent* if it is not the set of all formulas. Since L is closed under substitution, L is consistent iff no variable belongs to L .

We use the following logics (as sets of theorems of their usual Hilbert calculi, see [3]):

- **BCI**, **BCK**, **IL** and **CL** in the implicational language. **IL** = $\{K, S\}^*$ and **CL** = **IL** + P as in [14].
- **FL_e** (full Lambek calculus with exchange) and **FL_{ew}** (with exchange and weakening) in the language $\{\rightarrow, \cdot, \wedge, \vee, 0, 1\}$, possibly with $\perp, \top$. Intuitionistic logic in the full language contains **FL_{ew}**.

For implicational formulas, $\alpha \in \mathbf{CL}$ iff α is true under every Boolean valuation.

A *substitution* σ maps variables to formulas; $\sigma\alpha$ is the result of applying it to α . We write $\beta \leq_{\text{sub}} \alpha$ if $\alpha = \sigma\beta$ for some σ , and $\beta <_{\text{sub}} \alpha$ if $\beta \leq_{\text{sub}} \alpha$ but not $\alpha \leq_{\text{sub}} \beta$. A formula α is *minimal in L* if $\alpha \in L$ and there is no $\beta \in L$ with $\beta <_{\text{sub}} \alpha$. The *order* of an implicational formula is given by $\text{Ord}(p) = 1$ and $\text{Ord}(\alpha \rightarrow \beta) = \max\{\text{Ord}(\alpha) + 1, \text{Ord}(\beta)\}$.

Every logic L satisfies:

(C1) if $\gamma \in L$, then $\sigma\gamma \in L$ for every substitution σ ;

(C2) if $\gamma_1 \rightarrow \dots \rightarrow \gamma_k \rightarrow \beta \in L$ and $\gamma_1, \dots, \gamma_k \in L$, then $\beta \in L$.

Occurrences. We identify an occurrence in a formula α with its position in the syntax tree: a finite word over $\{0, 1\}$. The empty word ε is the root. If the subformula at u is $\gamma \circ \delta$ ($\circ$ a binary connective), then γ is at $u0$ and δ is at $u1$. We write α_u for the subformula at u . We say that u is *strictly below* v if v is a proper prefix of u . An occurrence u is a *leaf* if α_u is an atom, and a *compound occurrence* otherwise. Two occurrences of the same formula are never strictly below one another.

For a set Q of occurrences in α , no one of which is strictly below another, and a variable z , let $\alpha[Q := z]$ be the formula obtained from α by replacing the subformula at each $u \in Q$ by z .

3 The Polarized Statman Translation

Fix a formula α . Assign a *polarity* to each occurrence of α : ε is positive. If u is a compound occurrence with $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$, then $u1$ has the polarity of u and $u0$ the opposite one. If $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ with $\circ \in \{\cdot, \wedge, \vee\}$, both $u0$ and $u1$ have the polarity of u . For each compound occurrence u choose a new variable x_u (an *extension variable*), distinct from the variables of α and from each other. For every occurrence u put

$$r_u = \begin{cases} \alpha_u, & u \text{ a leaf,} \\ x_u, & u \text{ compound,} \end{cases}$$

and for every compound occurrence u with $\alpha_u = \alpha_{u0} \circ \alpha_{u1}$ put $b_u = r_{u0} \circ r_{u1}$ (the *body*) and

$$E_u = \begin{cases} b_u \rightarrow x_u, & u \text{ positive,} \\ x_u \rightarrow b_u, & u \text{ negative} \end{cases} \quad (\text{the link of } u).$$

Definition 3.1. Let $u_1, \dots, u_m$ be the compound occurrences of α in post-order. Then $S(\alpha) = E_{u_1} \rightarrow \dots \rightarrow E_{u_m} \rightarrow r_\varepsilon$.

For implicational α , $S(\alpha)$ is implicational. Jeřábek's φ^+ [4, Def. 3.6] uses $(b_u \rightarrow x_u) \wedge 1$ in place of E_u and a product in place of the iterated implication. We explain in Remark 6.2 why the “ $\wedge 1$ ” must be dropped here.

Example 3.2. Let $\alpha = ((a \rightarrow b) \rightarrow a) \rightarrow a$ (Peirce's law). Its compound occurrences are ε (positive), 0 (negative) and 00 (positive), with $\alpha_{00} = a \rightarrow b$ and $\alpha_0 = (a \rightarrow b) \rightarrow a$. Writing x, y, w for $x_\varepsilon, x_0, x_{00}$:

$$S(\alpha) = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (w \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Lemma 3.3. If α is implicational, $S(\alpha)$ has $3m + 1$ variable occurrences and $\text{Ord}(S(\alpha)) \leq 4$.

Proof. Each E_u has three variable occurrences. $\text{Ord}(b_u) = 2$, so $\text{Ord}(b_u \rightarrow x_u) = 3$ and $\text{Ord}(x_u \rightarrow b_u) = 2$. Hence $\text{Ord}(S(\alpha)) \leq \max\{3 + 1, 1\} = 4$. $\square$

The next lemma lists the facts about the shape of $S(\alpha)$ used below. All are immediate from the definition.

Lemma 3.4 (Soundness). *In each of the following cases, $L \vdash \alpha \rightarrow S(\alpha)$ for every formula α of the given language:*

- (i) $\mathcal{L} = \{\rightarrow\}$ and $L \supseteq \mathbf{BCI}$;
- (ii) $\mathcal{L} \subseteq \{\rightarrow, \cdot, 0, 1\}$ and $L \supseteq \mathbf{FL}_e$;
- (iii) $L \supseteq \mathbf{FL}_{ew}$, in any sublanguage of $\{\rightarrow, \cdot, \wedge, \vee, 0, 1, \perp, \top\}$ containing $\rightarrow$; in particular $L \supseteq \mathbf{IL}$.

Proof. We argue in the sequent calculus for $\mathbf{FL}_e$ (with weakening in case (iii)). For a compound occurrence u , let Γ_u be the multiset of the links E_v with $v = u$ or v strictly below u ; for a leaf, Γ_u is empty. We show by induction from the leaves upward:

$$\Gamma_u, \alpha_u \vdash r_u \text{ (} u \text{ positive),} \quad \Gamma_u, r_u \vdash \alpha_u \text{ (} u \text{ negative).}$$

For leaves this is an axiom. Let u be positive with $\alpha_u = \alpha_{u0} \rightarrow \alpha_{u1}$. The occurrence $u0$ is negative and $u1$ is positive. From the induction hypotheses and $\rightarrow$ -left, $\Gamma_{u0}, \Gamma_{u1}, \alpha_u, r_{u0} \vdash r_{u1}$. By $\rightarrow$ -right and $\rightarrow$ -left with $E_u = (r_{u0} \rightarrow r_{u1}) \rightarrow x_u$, we get $\Gamma_u, \alpha_u \vdash x_u$. The case of fusion is similar. Every formula of Γ_u is used exactly once, so no structural rule other than exchange is needed; this proves (i) and (ii). For $\wedge$ and $\vee$, the right- $\wedge$ and left- $\vee$ rules share the context. In each branch we remove the links of the other subtree by weakening; this proves (iii). Negative occurrences are symmetric. Taking $u = \varepsilon$ gives $E_{u_1}, \dots, E_{u_m}, \alpha \vdash r_\varepsilon$, i.e. $\vdash \alpha \rightarrow S(\alpha)$ after exchange. $\square$

Lemma 3.5 (Decoding). *Let θ_α be the substitution with $\theta_\alpha(x_u) = \alpha_u$ for each compound occurrence u , fixing all other variables. Then $\theta_\alpha(r_u) = \alpha_u$ for all u , and*

$$\theta_\alpha S(\alpha) = (\alpha_{u_1} \rightarrow \alpha_{u_1}) \rightarrow \dots \rightarrow (\alpha_{u_m} \rightarrow \alpha_{u_m}) \rightarrow \alpha.$$

Consequently, if a logic L proves all formulas $\gamma \rightarrow \gamma$, then $\alpha \in L + S(\alpha)$, and $S(\alpha) \in L$ implies $\alpha \in L$.

Proof. $\theta_\alpha(r_u) = \alpha_u$ holds by definition. Hence $\theta_\alpha(b_u) = \alpha_{u0} \circ \alpha_{u1} = \alpha_u = \theta_\alpha(x_u)$, so $\theta_\alpha(E_u) = \alpha_u \rightarrow \alpha_u$ in both polarities. The rest follows from (C1) and (C2). $\square$

Corollary 3.6. *In each case of Lemma 3.4, $L + \alpha = L + S(\alpha)$.*

4 Consequences for the Komori–Kashima Problem

In this section we derive the consequences stated in the introduction, assuming Theorem 1.1, which is proved in Section 5. We work in the implicational language.

Proof of Corollary 1.2. By Lemma 3.4, $L \vdash \alpha \rightarrow S(\alpha)$, so $S(\alpha) \in L$ whenever $\alpha \in L$. Apply Theorem 1.1. The equation $L + \alpha = L + S(\alpha)$ is Corollary 3.6. $\square$

Proof of Corollary 1.3. By Corollary 1.2(i) for L_0 , $L_0 + \alpha = L_0 + S(\alpha)$ for each $\alpha \in \Gamma$. By Theorem 1.1 with $L = \mathbf{CL}$ (where $S(\alpha) \in \mathbf{CL}$ by Lemma 3.4), each $S(\alpha)$ is minimal in $\mathbf{CL}$. It has order at most 4 by Lemma 3.3. $\square$

Proposition 4.1 (Four is optimal). *Every implicational formula in $\mathbf{CL}$ of order at most 3 is in $\mathbf{IL}$.*

Proof. A formula of order ≤ 2 has the form $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ with variables p_i, q ; read it as the rule “from $p_1, \dots, p_k$ infer q ”. A formula of order ≤ 3 is $F = H_1 \rightarrow \dots \rightarrow H_n \rightarrow r$ with such H_i and a variable r . Let C be the closure of $\emptyset$ under the rules $H_1, \dots, H_n$. The valuation making exactly C true satisfies every H_i , so $r \in C$ because $F \in \mathbf{CL}$. Following the closure step by step, modus ponens derives r from $H_1, \dots, H_n$ in $\mathbf{IL}$. Hence $F \in \mathbf{IL}$. $\square$

Corollary 4.2 (Answer to [14, Open Problem 2]). *Let G' be the formula of [14, Sect. 4]:*

$$G' = (a \rightarrow a') \rightarrow (b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \\ \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''')) \rightarrow c' \rightarrow c.$$

Then $S(G')$ is the formula

$$S(G') = (x_2 \rightarrow a \rightarrow a') \rightarrow (x_4 \rightarrow b' \rightarrow b) \rightarrow (x_6 \rightarrow c'' \rightarrow c') \rightarrow (x_8 \rightarrow c''' \rightarrow c') \\ \rightarrow ((a \rightarrow b) \rightarrow x_{11}) \rightarrow (x_{10} \rightarrow x_{11} \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow x_{14}) \rightarrow (x_{13} \rightarrow x_{14} \rightarrow c''') \\ \rightarrow (x_{18} \rightarrow a' \rightarrow b') \rightarrow ((x_{18} \rightarrow c'') \rightarrow x_{17}) \rightarrow (x_{21} \rightarrow b \rightarrow a) \rightarrow ((x_{21} \rightarrow c''') \rightarrow x_{20}) \\ \rightarrow (x_{22} \rightarrow c' \rightarrow c) \rightarrow (x_{19} \rightarrow x_{20} \rightarrow x_{22}) \rightarrow (x_{16} \rightarrow x_{17} \rightarrow x_{19}) \rightarrow ((x_{16} \rightarrow c) \rightarrow x_{15}) \\ \rightarrow ((x_{13} \rightarrow x_{15}) \rightarrow x_{12}) \rightarrow ((x_{10} \rightarrow x_{12}) \rightarrow x_9) \rightarrow ((x_8 \rightarrow x_9) \rightarrow x_7) \\ \rightarrow ((x_6 \rightarrow x_7) \rightarrow x_5) \rightarrow ((x_4 \rightarrow x_5) \rightarrow x_3) \rightarrow ((x_2 \rightarrow x_3) \rightarrow x_1) \rightarrow x_1.$$

It has 22 premises, 30 variables and 67 variable occurrences. It satisfies $\mathbf{IL} + S(G') = \mathbf{LC}$, it is minimal in $\mathbf{CL}$, and $\text{Ord}(S(G')) = 4$. Hence there exists a proper intermediate logic axiomatizable by a formula minimal in $\mathbf{CL}$ of order at most 4. That is, [14, Open Problem 2] has an affirmative answer, and the Komori–Kashima conjecture already fails for formulas of order at most 4.

Proof. By [14, Lemma 4.2], $\mathbf{IL} + G' = \mathbf{LC}$ and G' is minimal in $\mathbf{CL}$. Now apply Theorem 1.1, Corollary 3.6 and Lemma 3.3. The displayed formula is Definition 3.1, with the implication occurrences numbered as in Appendix A. $\square$

[Plot: Table 1 lists the subformula of G' that each x_i stands for. The Lean file checks that the displayed formula equals `statman Gprime 8` and has order 4.]

5 Proof of the Main Theorem

We follow the two steps described in the introduction. Section 5.1 proves the criterion (step 1). Section 5.2 proves the decoding lemma (step 2) and Theorem 1.1.

5.1 A criterion for minimality

Definition 5.1. Let δ be a formula, let Q be a nonempty set of occurrences of δ in α (that is, $\alpha_u = \delta$ for $u \in Q$), and let z be a variable not occurring in α . If δ is a variable, assume moreover that some occurrence of δ in α is not in Q . Then $\alpha[Q := z]$ is called a *one-step generalization* of α .

Example 5.2. For $\alpha = (a \rightarrow b) \rightarrow (a \rightarrow b)$, both $z \rightarrow (a \rightarrow b)$ (with $\delta = a \rightarrow b$) and $(z \rightarrow b) \rightarrow (a \rightarrow b)$ (with $\delta = a$) are one-step generalizations of α . $(z \rightarrow b) \rightarrow (z \rightarrow b)$ is not one, because all occurrences of a are replaced.

Lemma 5.3. *Every one-step generalization $\beta = \alpha[Q := z]$ of α satisfies $\beta <_{\text{sub}} \alpha$.*

Proof. $\beta \leq_{\text{sub}} \alpha$, because substituting δ for z in β gives α (z does not occur in α). Suppose, for a contradiction, that $\tau\alpha = \beta$ for some τ . Call connectives and constants *symbols*. A substitution never decreases the number of occurrences of symbols.

If δ is not a variable, it contains a symbol, so β has fewer symbols than α . This is a contradiction.

If δ is a variable p , then α and β have the same number of symbols, so τ maps each variable of α to a variable. Then $\tau\alpha$ has the same syntax tree as α , with the variable $\tau(p)$ at every leaf where α has p . But at these leaves β has z (at $u \in Q$) and p (at the occurrence outside Q). Hence $\tau(p) = z$ and $\tau(p) = p$, a contradiction. $\square$

Lemma 5.4. *Let L be a logic and $\alpha \in L$. If no one-step generalization of α is in L , then α is minimal in L .*

Proof. Let $\gamma \in L$ and $\sigma\gamma = \alpha$. We show $\alpha \leq_{\text{sub}} \gamma$.

Suppose first that σ maps the variables of γ injectively to variables. Then there is a substitution τ with $\tau(\sigma(q)) = q$ for every variable q of γ , and $\tau\alpha = \tau\sigma\gamma = \gamma$.

Otherwise, there is a variable q of γ such that either (i) $\sigma(q)$ is not a variable, or (ii) $\sigma(q) = \sigma(q')$ is a variable for some variable $q' \neq q$ of γ . Let z be a variable not occurring in α , and let σ' agree with σ except that $\sigma'(q) = z$. Every occurrence of q in γ yields an occurrence of $\delta := \sigma(q)$ in $\alpha = \sigma\gamma$. Let Q be the set of these occurrences; $Q \neq \emptyset$. Then $\sigma'\gamma = \alpha[Q := z]$. In case (ii), the occurrences of q' in γ yield occurrences of the variable δ that are not in Q . So $\sigma'\gamma$ is a one-step generalization of α , and $\sigma'\gamma \in L$ by (C1). This contradicts the hypothesis. $\square$

Corollary 5.5. *Let L be a logic. A formula $\alpha \in L$ is minimal in L if and only if no one-step generalization of α is in L .*

Proof. “If” is Lemma 5.4. “Only if” follows from Lemma 5.3. $\square$

5.2 Decoding one-step generalizations

The next lemma lists the facts about the shape of $S(\alpha)$ used in the proof. All of them are immediate from the definition.

Lemma 5.6. (a) *For each compound occurrence u , x_u occurs in $S(\alpha)$ exactly twice: once in E_u (its lower occurrence), and once as r_u (its upper occurrence). The upper occurrence lies inside $b_v \subseteq E_v$ for the parent v of u , or is the last formula r_ε when $u = \varepsilon$.*

(b) *For each atom c occurring in α , the occurrences of c in $S(\alpha)$ correspond bijectively to the leaves u of α with $\alpha_u = c$: the leaf u corresponds to the occurrence of $r_u = c$ in b_v for the parent v of u (or to the last formula when $\alpha = c$).*

(c) *Every compound subformula occurrence in $S(\alpha)$ is of exactly one of the following kinds: a body b_u inside E_u (both immediate subformulas are atoms); a link E_u (exactly one immediate subformula is an atom, the other is a body); a tail $T_i = E_{u_i} \rightarrow \dots \rightarrow E_{u_m} \rightarrow r_\varepsilon$ with $1 \leq i \leq m$ (its left immediate subformula is a link). In particular, a tail is not a subformula of a link.*

(d) *If $b_u = b_v$ as formulas, then $\alpha_u = \alpha_v$. If $E_u = E_v$ as formulas, then $u = v$.*

Proof. (a)–(c) follow by inspecting Definition 3.1. For the last claim of (c): the proper subformulas of a link are its body and atoms, and none of them is a tail. For (d), note that r_w determines α_w : if r_w is an atom of α , then $\alpha_w = r_w$; if $r_w = x_{w'}$, then $w = w'$. So $b_u = b_v$ gives the same main connective, $\alpha_{u0} = \alpha_{v0}$ and $\alpha_{u1} = \alpha_{v1}$, hence $\alpha_u = \alpha_v$. A positive link has the form (body) $\rightarrow$ (variable), and a negative one (variable) $\rightarrow$ (body). So $E_u = E_v$ forces equal polarity and $x_u = x_v$, hence $u = v$. $\square$

Idea of the proof. By Corollary 5.5, we must show that no one-step generalization α' of $S(\alpha)$ is in L . For each α' we find a substitution θ that turns every premise of α' into a formula of the form $\gamma \rightarrow \gamma$, and the conclusion into a one-step generalization β of α . By (C1) and (C2), $\alpha' \in L$ would give $\beta \in L$, which contradicts the minimality of α by Corollary 5.5. Lemma 3.5 is the special case where nothing is replaced. The next lemma provides the substitution in general.

Lemma 5.7 (Key lemma). *Let Q be a nonempty set of compound occurrences of α with $\alpha_v = \delta_0$ for all $v \in Q$, and let z be a variable occurring neither in α nor among the extension variables. Put $\beta = \alpha[Q := z]$; it is a one-step generalization of α . Call an occurrence u hidden if it is strictly below some $v \in Q$. For each occurrence u that is not hidden, β has an occurrence at u ; write β_u for its subformula ($\beta_u = z$ if $u \in Q$). Define θ by*

$$\theta(x_u) = \begin{cases} \alpha_u, & u \text{ hidden,} \\ \beta_u, & u \text{ not hidden,} \end{cases}$$

and let θ fix the variables of α . Then:

- (a) $\theta(r_u) = \alpha_u$ if u is hidden, and $\theta(r_u) = \beta_u$ otherwise;
- (b) for every compound occurrence $u \notin Q$, $\theta(E_u)$ has the form $\gamma \rightarrow \gamma$ (namely $\gamma = \theta(x_u)$);
- (c) for every $u \in Q$, $\theta(b_u) = \delta_0$ and $\theta(x_u) = z$;
- (d) $\theta(r_\varepsilon) = \beta$.

Proof. (a) For compound occurrences, this is the definition. A leaf u is not in Q , so $\beta_u = \alpha_u$ when u is not hidden, and $\theta(r_u) = \alpha_u$.

(b) Let $\circ$ be the main connective of α_u . If u is hidden, so are $u0$ and $u1$. By (a), $\theta(b_u) = \alpha_{u0} \circ \alpha_{u1} = \alpha_u = \theta(x_u)$. If u is not hidden and $u \notin Q$, then $u0$ and $u1$ are not hidden. Since β is not modified at u , $\theta(b_u) = \beta_{u0} \circ \beta_{u1} = \beta_u = \theta(x_u)$. In both cases $\theta(E_u)$ is $\gamma \rightarrow \gamma$ with $\gamma = \theta(x_u)$, whatever the polarity of u .

(c) If $u \in Q$, then $u0$ and $u1$ are hidden, so $\theta(b_u) = \alpha_{u0} \circ \alpha_{u1} = \alpha_u = \delta_0$. Also $\theta(x_u) = \beta_u = z$.

(d) ε is not hidden and $\beta_\varepsilon = \beta$. □

Proof of Theorem 1.1. Let $\alpha' = S(\alpha)[P := z]$ be a one-step generalization of $S(\alpha)$, where P is a set of occurrences of a formula δ . Assume $\alpha' \in L$; we derive a contradiction. By Lemma 5.6(c), δ falls under one of the following six cases.

Case 1: δ is a tail T_i . If $i = 1$, then $\alpha' = z$ is in L , contradicting consistency. Let $i \geq 2$. By Lemma 5.6(c), T_i does not occur inside $E_{u_1}, \dots, E_{u_{i-1}}$. Thus $\alpha' = E_{u_1} \rightarrow \dots \rightarrow E_{u_{i-1}} \rightarrow z$. By Lemma 3.5, $\theta_\alpha \alpha'$ consists of premises of the form $\gamma \rightarrow \gamma$ followed by z . By (C1) and (C2), $z \in L$, a contradiction.

Case 2: δ is a link E_u . By Lemma 5.6(c),(d), every occurrence of the formula E_u is the link occurrence E_u itself. So α' is $S(\alpha)$ with the premise E_u replaced by z . Take $Q = \{u\}$ and θ as in Lemma 5.7, and set $\theta(z) = z \rightarrow z$. Then $\theta \alpha'$ has premises $\theta(E_v)$ ($v \neq u$) of the form $\gamma \rightarrow \gamma$ by (b), the premise $z \rightarrow z$, and conclusion β by (d). By (C1) and (C2), $\beta \in L$.

Case 3: δ is a body b_u . By Lemma 5.6(c), each occurrence of δ is the body of some link. Hence P consists of the bodies of E_v for v in some nonempty set Q with $b_v = \delta$. By Lemma 5.6(d), all α_v ($v \in Q$) are the same formula δ_0 . Take θ as in Lemma 5.7, and set $\theta(z) = z$. For $v \in Q$, the premise E_v has become $z \rightarrow x_v$ or $x_v \rightarrow z$, and θ maps it to $z \rightarrow z$ by (c). The other premises and the conclusion are handled by (b) and (d). Hence $\beta \in L$.

Case 4: δ is an extension variable x_u . By Lemma 5.6(a), x_u has two occurrences. Since P is nonempty and does not contain both, P consists of exactly one of them. Take $Q = \{u\}$, $\delta_0 = \alpha_u$, and θ as in Lemma 5.7.

- (4a) P is the lower occurrence. Then E_u has become $b_u \rightarrow z$ or $z \rightarrow b_u$. Set $\theta(z) = \alpha_u$. By (c), this premise is mapped to $\alpha_u \rightarrow \alpha_u$. The rest is as before, so $\beta \in L$.
- (4b) P is the upper occurrence. Now E_u is unchanged, and x_u occurs in α' only in E_u . Modify θ by setting $\theta(x_u) = \alpha_u$ and $\theta(z) = z$. The formula z now sits exactly where x_u was read in $S(\alpha)$, and it is mapped to $z = \beta_u$ as before. So (b) and (d) still hold for all links other than E_u . Moreover $\theta(E_u)$ is $\alpha_u \rightarrow \alpha_u$ by (c). Hence $\beta \in L$.

Case 5: δ is a variable p of α . By Lemma 5.6(b), P corresponds to a set Q of leaves of α labelled p . Q is nonempty and does not contain all of them, so $\beta = \alpha[Q := z]$ is a one-step

generalization of α . Since $S(\cdot)$ is computed leaf by leaf on the same syntax tree, $\alpha' = S(\beta)$ (with the same extension variables). By Lemma 3.5, $S(\beta) \in L$ implies $\beta \in L$.

Case 6: δ is a constant c . Exactly as in Case 5: $\alpha' = S(\beta)$ with $\beta = \alpha[Q := z]$ for a nonempty set Q of leaves labelled c . β is a one-step generalization, because c is not a variable. Hence $\beta \in L$.

In Cases 2–6 we obtained $\beta \in L$ for a one-step generalization β of α . This contradicts the minimality of α by Corollary 5.5. Hence no one-step generalization of $S(\alpha)$ is in L . Since $S(\alpha) \in L$ by assumption, $S(\alpha)$ is minimal by Corollary 5.5. $\square$

Example 5.8. We illustrate Case (4b) with Example 3.2. Replace the upper occurrence of $w = x_{00}$ by z :

$$\alpha' = ((a \rightarrow b) \rightarrow w) \rightarrow (y \rightarrow (z \rightarrow a)) \rightarrow ((y \rightarrow a) \rightarrow x) \rightarrow x.$$

Here $\beta = \alpha[\{00\} := z] = (z \rightarrow a) \rightarrow a$. The substitution is $w \mapsto a \rightarrow b$, $y \mapsto z \rightarrow a$, $x \mapsto (z \rightarrow a) \rightarrow a$, $z \mapsto z$:

$$\theta\alpha' = ((a \rightarrow b) \rightarrow (a \rightarrow b)) \rightarrow ((z \rightarrow a) \rightarrow (z \rightarrow a)) \rightarrow (\beta \rightarrow \beta) \rightarrow \beta.$$

Since $\beta \notin \mathbf{CL}$ ($z = 0$, $a = 0$), neither is α' .

6 Further Remarks

This section collects supplementary results. They are not needed for the main line of the paper.

6.1 How far can the base logic be weakened?

Theorem 1.1 needs only $\gamma \rightarrow \gamma \in L$, consistency, and $S(\alpha) \in L$. In the implicational case, the last condition is guaranteed by **BCI** (Lemma 3.4). Neither B nor C can be dropped. We write **XY**... for the implicational logic axiomatized by the axiom schemes $X, Y, \dots$ among

$$B = (b \rightarrow c) \rightarrow (a \rightarrow b) \rightarrow a \rightarrow c, \quad B' = (a \rightarrow b) \rightarrow (b \rightarrow c) \rightarrow a \rightarrow c, \quad C = (a \rightarrow b \rightarrow c) \rightarrow b \rightarrow a \rightarrow c,$$

$K = a \rightarrow b \rightarrow a$ and $I = a \rightarrow a$, with modus ponens and substitution.

Proposition 6.1. (a) (Without B .) K is minimal in **CK**, but $S(K) = ((b \rightarrow a) \rightarrow x_1) \rightarrow ((a \rightarrow x_1) \rightarrow x_0) \rightarrow x_0 \notin \mathbf{CK}$.

(b) (Without C .) I is minimal in **BB'KI**, but $S(I) = ((a \rightarrow a) \rightarrow x) \rightarrow x \notin \mathbf{BB'KI}$. Likewise K is minimal in **BK** and in **BB'KI**, but $S(K) \notin \mathbf{BB'KI} \supseteq \mathbf{BK}$.

Hence, over **CK**, **BK** and **BB'KI** (and every logic between **BK** and **BB'KI**), S does not preserve minimality, and $L + \alpha \neq L + S(\alpha)$ for $\alpha = K$ (resp. I).

Proof. Minimality. The one-step generalizations of K are $z \rightarrow b \rightarrow a$, $a \rightarrow b \rightarrow z$, $a \rightarrow z$ and z ; those of I are $z \rightarrow a$, $a \rightarrow z$ and z . None of them is classically valid, so none lies in any of the logics considered, all of which are contained in **CL**.

(a) Let $M_4 = (\{0, 1, 2, 3\}, \rightarrow, \{0\})$ with the table

$\rightarrow$	0	1	2	3
0	0	1	2	3
1	0	0	1	0
2	0	0	0	3
3	0	1	2	0

A finite check shows that M_4 validates C and K (hence also I , which is derivable in **CK**), but not B . The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ forces $y = 0$. So every **CK**-theorem takes the value 0. With $a = b = 1$, $x_1 = 2$ and $x_0 = 3$, however, $b \rightarrow a = 0$, $(b \rightarrow a) \rightarrow x_1 = 2$, $a \rightarrow x_1 = 1$ and $(a \rightarrow x_1) \rightarrow x_0 = 0$. Hence $S(K)$ takes the value $2 \rightarrow (0 \rightarrow 3) = 2 \rightarrow 3 = 3 \neq 0$. Intuitively, $S(K)$ needs the composition $\lambda y. e_1(\lambda z. y)$ inside an argument, which the combinators **C**, **K** cannot form. By contrast, $S(I) = ((a \rightarrow a) \rightarrow x) \rightarrow x$ is a **CK**-theorem (it is the type of **CII**).

(b) Let $M_3 = (\{0, 1, 2\}, \rightarrow, \{0\})$ with $x \rightarrow y = 0$ if $y \leq x$ and $x \rightarrow y = 1$ otherwise. M_3 validates B, B', K, I : this is a finite check. The set $\{0\}$ is closed under modus ponens, since $0 \rightarrow y = 0$ forces $y = 0$. So every theorem of **BB'KI** takes the value 0 in M_3 . With $a = 0$ and $x = x_0 = 2$, however, $S(I)$ takes the value $(0 \rightarrow 2) \rightarrow 2 = 1 \rightarrow 2 = 1$. $S(K)$ is refuted similarly. (M_3 refutes C and W .) $\square$

[Plot: Checks: `tools/ck_z3.py` (matrix M_4 , found with `z3` and verified by exhaustive evaluation), `tools/bki_matrix.py` (matrix M_3), `tools/ck_cd.py` and `tools/bk_cd.py` (condensed-detachment search, consistent with (a) and (b)). Open: ticket entailment $\mathbf{T}_{\rightarrow} = \mathbf{BB'IW}$, where M_3 does not apply because it refutes W ; and whether reordering the links helps. For (b), every order of links fails already for $S(I)$, which has a single link.]

6.2 Additive connectives over $\mathbf{FL}_e$

Remark 6.2 (Additive connectives over $\mathbf{FL}_e$). The proof of Theorem 1.1 uses only (C1), (C2), $\gamma \rightarrow \gamma \in L$ and consistency. Polarity is used only in Lemma 3.4, which is where $\mathbf{FL}_e$ with $\wedge, \vee$ fails. Two facts, both checked with a cut-free prover for $\mathbf{FL}_e$ (`tools/fle_check.py`), show that the problem cannot be removed by simply adding Jeřábek's " $\wedge 1$ ":

- (a) *Without wrappers, S need not be sound.* $\alpha = (q \rightarrow q) \wedge (r \rightarrow r)$ is minimal in $\mathbf{FL}_e$, but $S(\alpha) = ((q \rightarrow q) \rightarrow x_0) \rightarrow ((r \rightarrow r) \rightarrow x_1) \rightarrow ((x_0 \wedge x_1) \rightarrow x) \rightarrow x$ is not provable in $\mathbf{FL}_e$: each branch of the $\wedge$ -right rule would have to discard the other branch's link.
- (b) *With wrappers, minimality is always lost.* Jeřábek's φ^+ puts $(E_u \wedge 1)$ in place of E_u . The link of the root is never discarded, so replacing its 1 by z gives a one-step generalization that is still provable (by $\wedge$ -left). Hence φ^+ is not minimal in any $L \supseteq \mathbf{FL}_e$ (including $\mathbf{FL}_{ew}$, $\mathbf{IL}$, $\mathbf{CL}$) whenever φ is compound. For example, $((p \rightarrow p) \rightarrow x) \wedge z \rightarrow x$ is provable.

Exhaustive tests over all formulas with at most four binary connectives in $\rightarrow, \wedge, \vee$ and three variables give the following. There are 413 formulas minimal in $\mathbf{FL}_e$, up to renaming. For 164 of them $S(\alpha)$ is provable, and in all 164 cases it is minimal, as Theorem 1.1 predicts. φ^+ is provable in all 413 cases and minimal in none. Wrapping only the links strictly below an

additive occurrence gives a provable formula in all 413 cases, but a minimal one in only 255. For $\alpha = p \vee (q \rightarrow q)$, for instance, the wrapper can be removed because the $\vee$ -right rule selects one branch. So a minimality-preserving translation over $\mathbf{FL}_e$, if one exists, must adapt the wrappers to the actual proof. Whether such a translation exists, and the non-commutative case $\mathbf{FL}$, are left open.

6.3 Remarks on the proofs

Remark 6.3. Corollary 5.5 replaces [14, Props. 3.1–3.2], whose proofs there are sketched, and it does not need simple formulas.

Remark 6.4. In Lemma 3.4(i), linear λ -terms can be given instead of sequent derivations. Without exchange the premises of $S(\alpha)$ cannot be reordered; see Section 6.1.

6.4 Top-level premises of order 3

For $F = A_1 \rightarrow \cdots \rightarrow A_n \rightarrow r$ with r a variable, we call $A_1, \dots, A_n$ the *top-level premises* of F . The next proposition shows that an order-4 counter-example needs at least two top-level premises of order 3. This explains, in part, why $S(G')$ is long.

Proposition 6.5. *Let $F = A_1 \rightarrow \cdots \rightarrow A_n \rightarrow r \in \mathbf{CL}$ with r a variable and $\text{Ord}(F) \leq 4$. If at most one A_i has order 3, then $F \in \mathbf{IL}$ or $\mathbf{IL} + F = \mathbf{CL}$. Minimality of F is not assumed.*

Proof. A formula of order ≤ 2 has the form $p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$ with variables p_i, q ($k \geq 0$). We call it a *Horn clause* and read it as the rule “from $p_1, \dots, p_k$ infer q ”. Since $\text{Ord}(F) \leq 4$, every A_i has order ≤ 3 . Recall that premises can be permuted in $\mathbf{IL}$.

Let Γ be the set of top-level premises of order ≤ 2 . For a set X of variables, let $C(X)$ be the closure of X under the rules in Γ . We identify a set of variables with the valuation making exactly those variables true. Then:

(H1) $C(X)$ satisfies Γ ;

(H2) if $q \in C(X)$, then $\Gamma, X \vdash_{\mathbf{IL}} q$ (modus ponens along the closure). In particular, $q \in C(\{p_1, \dots, p_k\})$ implies $\Gamma \vdash_{\mathbf{IL}} p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$.

No premise has order 3. Then F is $\Gamma \rightarrow r$. By (H1) and $F \in \mathbf{CL}$, we get $r \in C(\emptyset)$, so $F \in \mathbf{IL}$ by (H2).

Exactly one premise, $A = B_1 \rightarrow \cdots \rightarrow B_m \rightarrow s$, has order 3. Then each B_j has order ≤ 2 . Assume $F \notin \mathbf{IL}$. Put $C_0 = C(\emptyset)$.

- $r \notin C_0$, since otherwise $\Gamma \vdash_{\mathbf{IL}} r$ by (H2).
- Some B_j is not derivable from Γ in $\mathbf{IL}$. Otherwise $\Gamma, A \vdash_{\mathbf{IL}} s$. Moreover, the valuation $C(\{s\})$ satisfies Γ and A , so $r \in C(\{s\})$ because $F \in \mathbf{CL}$. Then $\Gamma, s \vdash_{\mathbf{IL}} r$ by (H2), and $F \in \mathbf{IL}$.

Fix such a $B_j = p_1 \rightarrow \cdots \rightarrow p_k \rightarrow q$, and put $C_1 = C(\{p_1, \dots, p_k\})$. Then $C_0 \subseteq C_1$, and $q \notin C_1$ by (H2). The valuation C_1 falsifies B_j , hence satisfies A . It also satisfies Γ by (H1). Since $F \in \mathbf{CL}$, we get $r \in C_1$.

Let $H_3 = \{0, \frac{1}{2}, 1\}$ be the three-element Gödel algebra, with $x \Rightarrow y = 1$ if $x \leq y$ and $x \Rightarrow y = y$ otherwise. Define $v(t) = 1$ if $t \in C_0$, $v(t) = \frac{1}{2}$ if $t \in C_1 \setminus C_0$, and $v(t) = 0$ otherwise. A Horn clause $p_1 \rightarrow \dots \rightarrow p_k \rightarrow q$ has value 1 under v iff $\min_i v(p_i) \leq v(q)$ (the minimum of an empty set is 1).

- Every clause in Γ has value 1, since C_0 and C_1 are closed under the rules in Γ .
- B_j has value 0, since $v(p_i) \geq \frac{1}{2}$ for all i and $v(q) = 0$. Hence A has value 1.
- $v(r) = \frac{1}{2}$, since $r \in C_1 \setminus C_0$.

So F has value $\frac{1}{2}$, and $H_3 \not\models F$. By Jankov's criterion [11, Theorem 1] (see [14, Sect. 3.2]), an implicational $F \in \mathbf{CL} \setminus \mathbf{IL}$ with $H_3 \not\models F$ satisfies $\mathbf{IL} + F = \mathbf{CL}$. $\square$

Remark 6.6. In $\mathbf{S}(\alpha)$, a positive link $b_u \rightarrow x_u$ has order 3 and a negative link $x_u \rightarrow b_u$ has order 2. So the top-level premises of order 3 in $\mathbf{S}(\alpha)$ are exactly the links of the positive implication occurrences of α . Hence, by Proposition 6.5, if $\alpha \in \mathbf{CL}$ has at most one positive implication occurrence, then $\mathbf{IL} + \alpha = \mathbf{IL} + \mathbf{S}(\alpha)$ is $\mathbf{IL}$ or $\mathbf{CL}$.

$\mathbf{S}(G')$ has 11 positive links: $E_1, E_3, E_5, E_7, E_9, E_{11}, E_{12}, E_{14}, E_{15}, E_{17}, E_{20}$. By Proposition 6.5, a $\mathbf{CL}$ -minimal axiom of order 4 for a proper intermediate logic has at least two top-level premises of order 3. The least possible number is open.

7 Formalization

[Plot: Half a page.]

File `lean/StatmanMinimality.lean` (about 1300 lines, implicational language): Lean 4 core only, no Mathlib, no `sorry`, no `native_decide`. `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. Theorem 1.1 is formalized in exactly the generality stated: a logic is a predicate `Th` with `Logic Th` (closed under substitution and modus ponens, proving $\gamma \rightarrow \gamma$, consistent), and `statman_minimalIn : Logic Th  $\rightarrow$  ( $\forall q, \text{Occ } q \text{ A} \rightarrow q < N$ )  $\rightarrow$  MinimalIn Th A  $\rightarrow$  Th (statman A N)  $\rightarrow$  MinimalIn Th (statman A N)`. The classical instance `statman_minimal` uses `logic_valid` and `statman_valid`. Lemma 3.4(i) and Corollary 3.6 are formalized for every implicational logic containing **BCI** (`BCIExt Th`: closed under substitution and modus ponens, proving all instances of B, C, I): `BCIExt.statman_imp : Th (A  $\rightarrow$  statman A N), BCIExt.statman_iff : Th A  $\leftrightarrow$  Th (statman A N)`, and `BCIExt.statman_min` (Corollary 1.2 in the implicational language, without extra hypotheses). Also formalized: `statman_ord` (order ≤ 4) and `statman_gprime` (Corollary 4.2: the displayed $\mathbf{S}(G')$ equals `statman gprime 8`). Not formalized: the full language (Lemma 3.4(ii),(iii)). Correspondence: `reduction` and `strict` = Corollary 5.5; `decode/decode_node` = Lemma 5.7 and Cases 2–5; the tail branch of `statman_minimalIn` = Case 1.

8 Concluding Remarks

We showed that the polarized Statman translation preserves minimality with respect to the substitution preorder in any consistent logic that proves $\gamma \rightarrow \gamma$ and $\mathbf{S}(\alpha)$. As a consequence, $\mathbf{CL}$ -minimal axiomatizations of implicational logics over **BCI** can be taken of order 4, and [14, Open Problem 2] has an affirmative answer.

Open problems:

1. Which intermediate logics are axiomatizable by **CL**-minimal formulas [14, Open Problem 1]? By Corollary 1.3 it suffices to consider formulas of order 4.
2. Is there a minimality-preserving translation over **FL_e** with additive connectives, or over the non-commutative **FL** (Section 6.2)?
3. Does the translation (possibly with reordered premises) preserve minimality over ticket entailment $\mathbf{T}_{\rightarrow} = \mathbf{BB'IW}$ (Section 6.1)?
4. What is the least number of top-level premises of order 3 in a **CL**-minimal order-4 axiom of a proper intermediate logic? It is at least 2 (Proposition 6.5), and $\mathbf{S}(G')$ has 11.

Acknowledgements. [\[Plot: JSPS KAKENHI etc.\]](#)

A The formula $\mathbf{S}(G')$

The implication occurrences of G' are numbered $1, \dots, 22$ in pre-order, and x_i is the extension variable of occurrence i . $\mathbf{S}(G')$ lists the links in the order of the table (post-order), followed by x_1 . The last column gives the subformula G'_u at occurrence i , so that the decoding substitution of Lemma 3.5 is $x_i \mapsto G'_u$. The table is generated by `paper/gen_SGprime.py`.

Table 1: The links of $\mathbf{S}(G')$.

i	pol.	E_i	G'_u
E_2	—	$x_2 \rightarrow a \rightarrow a'$	$a \rightarrow a'$
E_4	—	$x_4 \rightarrow b' \rightarrow b$	$b' \rightarrow b$
E_6	—	$x_6 \rightarrow c'' \rightarrow c'$	$c'' \rightarrow c'$
E_8	—	$x_8 \rightarrow c''' \rightarrow c'$	$c''' \rightarrow c'$
E_{11}	+	$(a \rightarrow b) \rightarrow x_{11}$	$a \rightarrow b$
E_{10}	—	$x_{10} \rightarrow x_{11} \rightarrow c''$	$(a \rightarrow b) \rightarrow c''$
E_{14}	+	$(b' \rightarrow a') \rightarrow x_{14}$	$b' \rightarrow a'$
E_{13}	—	$x_{13} \rightarrow x_{14} \rightarrow c'''$	$(b' \rightarrow a') \rightarrow c'''$
E_{18}	—	$x_{18} \rightarrow a' \rightarrow b'$	$a' \rightarrow b'$
E_{17}	+	$(x_{18} \rightarrow c'') \rightarrow x_{17}$	$(a' \rightarrow b') \rightarrow c''$
E_{21}	—	$x_{21} \rightarrow b \rightarrow a$	$b \rightarrow a$
E_{20}	+	$(x_{21} \rightarrow c''') \rightarrow x_{20}$	$(b \rightarrow a) \rightarrow c'''$
E_{22}	—	$x_{22} \rightarrow c' \rightarrow c$	$c' \rightarrow c$
E_{19}	—	$x_{19} \rightarrow x_{20} \rightarrow x_{22}$	$((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$
E_{16}	—	$x_{16} \rightarrow x_{17} \rightarrow x_{19}$	$((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$
E_{15}	+	$(x_{16} \rightarrow c) \rightarrow x_{15}$	$((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c$
E_{12}	+	$(x_{13} \rightarrow x_{15}) \rightarrow x_{12}$	$((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_9	+	$(x_{10} \rightarrow x_{12}) \rightarrow x_9$	$((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$

i	pol.	E_i	G'_u
E_7	+	$(x_8 \rightarrow x_9) \rightarrow x_7$	$(c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow$ $((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow$ $c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow$ c
E_5	+	$(x_6 \rightarrow x_7) \rightarrow x_5$	$(c'' \rightarrow c') \rightarrow (c''' \rightarrow c') \rightarrow ((a \rightarrow$ $b) \rightarrow c'') \rightarrow ((b' \rightarrow a') \rightarrow c''') \rightarrow$ $((a' \rightarrow b') \rightarrow c'') \rightarrow ((b \rightarrow a) \rightarrow$ $c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_3	+	$(x_4 \rightarrow x_5) \rightarrow x_3$	$(b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow (c''' \rightarrow$ $c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow ((b' \rightarrow$ $a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow c'') \rightarrow$ $((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow c$
E_1	+	$(x_2 \rightarrow x_3) \rightarrow x_1$	$(a \rightarrow a') \rightarrow (b' \rightarrow b) \rightarrow (c'' \rightarrow c') \rightarrow$ $(c''' \rightarrow c') \rightarrow ((a \rightarrow b) \rightarrow c'') \rightarrow$ $((b' \rightarrow a') \rightarrow c''') \rightarrow (((a' \rightarrow b') \rightarrow$ $c'') \rightarrow ((b \rightarrow a) \rightarrow c''') \rightarrow c' \rightarrow c) \rightarrow$ c

References

- [1] Barendregt, H., W. Dekkers, and R. Statman, *Lambda Calculus with Types*, Cambridge University Press, 2013.
- [2] Ciabattoni, A., N. Galatos, and K. Terui, From axioms to analytic rules in nonclassical logics, in *LICS 2008*, IEEE, 2008, pp. 229–240.
- [3] Galatos, N., P. Jipsen, T. Kowalski, and H. Ono, *Residuated Lattices: An Algebraic Glimpse at Substructural Logics*, Elsevier, 2007.
- [4] Jeřábek, E., A note on the substructural hierarchy, *Mathematical Logic Quarterly* 62(1–2):102–110, 2016. arXiv:1507.00700.
- [5] Mints, G., Gentzen-type systems and resolution rules. Part I: propositional logic, in *COLOG-88*, LNCS 417, Springer, 1990, pp. 198–231.
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- [6] Plaisted, D. A., and S. Greenbaum, A structure-preserving clause form translation, *Journal of Symbolic Computation* 2(3):293–304, 1986.
- [7] Rybakov, V. V., *Admissibility of Logical Inference Rules*, Elsevier, 1997.
- [8] Švejdar, V., On the polynomial-space completeness of intuitionistic propositional logic, *Archive for Mathematical Logic* 42(7):711–716, 2003.
- [9] Tseitin, G. S., On the complexity of derivation in propositional calculus, in *Studies in Constructive Mathematics and Mathematical Logic, Part II*, 1968, pp. 115–125.
- [10] Dummett, M., A propositional calculus with denumerable matrix, *The Journal of Symbolic Logic* 24(2):97–106, 1959.

- [11] Jankov, V. A., On the extension of the intuitionist propositional calculus to the classical calculus, and the minimal calculus to the intuitionist calculus, *Mathematics of the USSR-Izvestiya* 2(1):205–208, 1968.
- [12] Kashima, R., Problems on axiomatization of intermediate propositional logics, in *the 39th MLG meeting*, 2005, pp. 59–62.
- [13] Komori, Y., A problem on logics axiomatized with formulas minimal in classical logic, and more (in Japanese), MSJ Autumn Meeting, 2005.
- [14] Nakamura, Y., and N. Matsuda, On implicational intermediate logics axiomatizable by formulas minimal in classical logic: a counter-example to the Komori–Kashima problem, *Studia Logica* 109:1413–1422, 2021.
- [15] Statman, R., Intuitionistic propositional logic is polynomial-space complete, *Theoretical Computer Science* 9:67–72, 1979.
- [16] de Moura, L., and S. Ullrich, The Lean 4 theorem prover and programming language, in *CADE-28*, LNCS 12699, 2021, pp. 625–635.